Chapter 13, Problem 77P

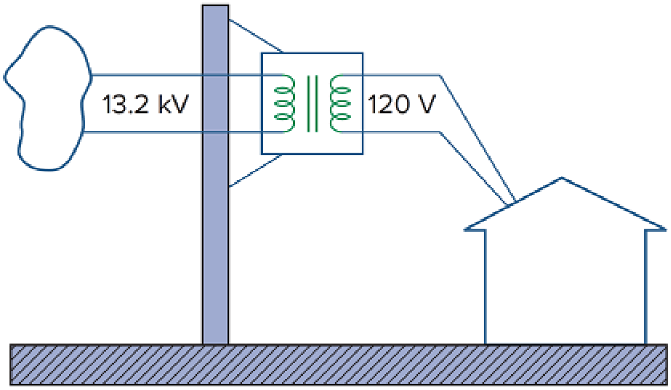
Problem

The three-phase system of a town distributes power with a line voltage of 13.2 kV. A pole transformer connected to single wire and ground steps down the high-voltage wire to 120 V rms and serves a house as shown in Fig. 13.139.

(a) Calculate the turns ratio of the pole transformer to get 120 V.

(b) Determine how much current a 100-W lamp connected to the 120-V hot line draws from the high-voltage line.

Figure 13.139



Step-by-step solution

Step 1 of 2

(A) This is a single phase transformer,

$$V_1 = 13.2\ KV\ ,\quad V_2 = 120\ V$$

$$n = \frac{V_2}{V_1} = \frac{120}{13200} = \frac{1}{110}$$

Therefore,

$$n = \frac{1}{110}$$

Or,

110 turn on the primary to every turn on the secondary.

Step 2 of 2

(B) $P = V\,I$

Or,

$$I = \frac{P}{V} = \frac{100}{120} = 0.833\ A$$

$$I_1 = n I_2 = \frac{0.8333}{110} = 7.7576\ mA\ .$$